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Master Thesis

GRAPH NEURAL NETWORKS FOR MONEY LAUNDERING DETECTION

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ABSTRACT

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INTRODUCTION

Money laundering (ML) is the criminal process of disguising or concealing the origin, identity, and ownership of illegally acquired funds. Its detection has become a foundational pillar of modern economic security. The rapid digitization of financial services has opened new possibilities for fraudulent activities. Undetected ML leads to massive economic losses for individuals and institutions, undermines the public trust in the financial system and frequently serves to finance criminal organizations. The scale of this phenomenon is staggering: the United Nations Office on Drugs and Crime (UNODC) estimates that between 2% and 5% of global GDP is laundered annually, with less than 1% of these illicit financial flows being successfully detected [15, 9]. According to the Canadian Anti Fraud Centre (CAFC), 704 million dollars were lost because of frauds in 2025 only, confirming the raising trend of the previous years [3]. Consequently, developing robust mechanisms to identify and stop it is critically important to preserving both market integrity and social stability.

1.1 TRADITIONAL AML METHODS

Anti Money Laundering is a critical component of the global financial system. Financial institutions ought to implement robust mechanisms to comply with regulatory standards and protect the integrity of the financial system [12].

Rule-based models have long been employed to detect suspicious transactions. These models consist of a set of predefined rules that are used to flag transactions that deviate from typical financial behavior. The criteria are typically based on regulatory requirements, historical data, and known patterns of illicit behavior. For example we could decide to flag transactions that exceed a certain amount, that rapidly transfer funds between accounts or that involve a high-risk jurisdiction.

The main advantage of this approach is its transparency and interpretability. Furthermore they are computationally inexpensive, easy to implement and they can operate in real-time [12]. Unfortunately there are also several limitations. They tend to produce a high amount of false positives, which leads to more work from human operators that will have to deal with the flagged transactions. They could be easily exploited by criminals who are aware of the rules and can tailor their illicit activities to avoid detection. This will be further discussed in Chapter 2.

Due to its tendency of producing a huge amount of false positives and the huge amount of data that nowadays is produced by financial institutions, the rule-based approach is becoming less and less effective.

At the same time big datasets allow the usage of data-driven solutions that can learn directly from historical data. First attempts in this direction involved standard machine learning approaches such as logistic regression and support vector machines [4]. Those models improved the state of the art but they still struggled to detect human brain-armed patterns due to their very limited parameter capacity [5].

For this reason the research field shifted toward deep learning solutions which offer a much wider and complex hypothesis space, allowing to capture high order relationships between data, at the cost of a much worse interpretability. Nowadays the challenge is to combine high performance together with explainability to allow the usage of effective models in the highly regulated financial field, a necessity increasingly emphasized by both financial institutions and supervisory authorities [10]. We will discuss this further in Chapter 3

1.2 THE GROWING CHALLENGE

Detecting fraud is becoming increasingly difficult. The sheer volume and velocity of digital transactions make manual oversight impossible and heavily strain classical machine learning architectures. Furthermore, the rise of decentralized finance (DeFi), Web3, and cryptocurrencies has introduced a paradigm of pseudo-anonymity that complicates traditional tracing.

On top of that, criminals are constantly developing more and more sophisticated methods to avoid being caught. As Cuèllar (2003) noted, *"raising the marginal cost of perpetrating criminal activity can itself conceivably have counterintuitive effects. To the extent that criminals engaged in crimes for profit are acting rationally, then raising the cost of crime can discourage smaller operations and clear the market allowing more sophisticated criminal* *organizations to innovate and take more control from less organized criminals"* [8]. Hence, to prevent inadvertently incentivizing sophisticated criminal networks, detection mechanisms must be updated continuously. The framework proposed in this study leverages deep learning models that support online updates. This ensures that the detection algorithm remains aligned with the most recent transaction patterns.

1.3 GNN FOR FRAUD DETECTION

The necessity of capturing complex interactions between accounts and transactions led to the idea of modeling the financial feed as a graph, where accounts are nodes and transactions are edges [13].

Definition 1.1 (Financial Graph)**.** *A financial graph is a directed graph* $G = (V, E)$*, where* V *represents a set of financial entities (e.g. bank accounts or individuals), and* E *represents a set of transactions between these entities. Each edge* $e \in E$ *is directed from the sender to the receiver and may be associated with features.*

GNNs rely on message passing to aggregate information from neighbors and learn representations of the nodes that will be given as input to a classifier or regressor to predict the probability that an account (Node-Level detection), a transaction (Edge-Level detection) or a subgraph (Graph-Level detection) is fraudulent. These methods allow to capture complex relationships, learn features automatically, handle dynamic graphs, and integrate multimodal data.

Formally the message passing process at the k-th layer in a GNN is defined as follows:

$$\begin{aligned} m_i^{(k)} &= \text{AGGREGATE}\left\{\text{MSG}\left(h_j^{(k-1)}, e_{ji}\right) : j \in \mathcal{N}(i)\right\} \\ h_i^{(k)} &= \text{UPDATE}\left(h_i^{(k-1)}, m_i^{(k)}, x_i\right) \end{aligned} \tag{1.1}$$

where $\mathcal{N}(i)$ denotes the set of neighboring nodes of node i, e_{ji} is the feature vector of the edge from node j to node i, MSG (e.g. MLP) is the function that merges informations from the neighbor and the connecting edge and x_i is the feature vector of node i. Sometimes edge features are not used during the message passing process, in such cases $m_i^{(k)}$ is defined as:

$$m_i^{(k)} = \text{AGGREGATE}\left\{h_j^{(k-1)} : j \in \mathcal{N}(i)\right\} \tag{1.2}$$

After K iterations of message passing, the resultant node embeddings $H^{(K)}$ are then used to execute the designated task, in our case the detection of fraudulent accounts [6].

There are different types of GNNs; they are defined by how the UPDATE and AGGREGATE functions are implemented. Further details will be provided in Chapter 5.

1.4 STRUCTURE OF THE DISSERTATION

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FINANCIAL CRIME AND MONEY LAUNDERING

Understanding how money laundering works is fundamental to garantee both an effective and explainable detection of suspicious transaction. This chapter offers a theoretical introduction to money laundering providing a mathematical formulation of the problem together with a summary of common strategies employed by criminals to avoid detection.

2.1 THE LIMITATIONS OF THE LONELY SCAMMER

Individual scammers typically rely on simple and intuitive techniques to introduce illicit funds into the legitimate economy. One of the most prevalent methods is known as “smurfing”. In simple terms, smurfing involves breaking down a large sum of illicit money into many smaller, inconspicuous chunks to deposit them without raising suspicion.

More formally, individual scammers rely on smurfing to avoid Anti-Money Laundering (AML) detection mechanisms, which usually flag transactions above a specific threshold $\tilde{x}$. By splitting their initial illicit funds x into smaller transactions x_n, such that each transaction satisfies $x_n \leqslant \tilde{x}$, they minimize the risk of detection [7]. The mathematical expectation of profit π_s for a scammer engaging in smurfing can be defined as:

$$\pi_s = x(1 - \nu) \tag{2.1}$$

where x represents the total initial illicit funds, and $\nu \in (0, 1)$ is the friction cost or commission proportion associated with the smurfing process (e.g., the fee paid to mules).

While smurfing is effective for small amounts, its scalability is severely limited. When attempting to launder much larger volumes of funds, an individual scammer operating from a single account faces a significantly higher probability of detection $(1 - p)$, where p is the probability of successfully avoiding detection. If caught, the scammer is subjected to a severe monetary penalty T, which includes both the total confiscation of the funds and associated judicial fines [7]. Therefore, operating as a "lonely scammer" becomes mathematically and practically unviable for large-scale operations due to the massive expected cost of $(1-p)T$.

2.2 FRAUD RINGS

In regard to overcome those issues, criminals have improved their techniques by employing highly sophisticated obfuscation tactics. Instead of acting alone, they form complex "fraud rings" to distribute illicit funds across vast networks of temporary accounts. To further break the traceable link between sender and receiver, illicit actors frequently use cyclic transfers and cryptocurrency **mixers** services that pool together funds from multiple users, mix them, and redistribute them at random intervals and amounts. These advanced tactics create dense, highly camouflaged transaction topologies that easily evade threshold-based rules and simple heuristic detection methods.

To achive their goal they utilize a strategy of transaction layering, often characterized by a topological depth L, and network diversification to obscure the money trail [7, 11]. By spreading the funds through a complex topology, the fraud ring successfully increases its non-detection probability p_L, which now depends explicitly on the layering depth L. The expected profit of a fraud ring utilizing L layers can be mathematically expressed as:

$$E[\pi_L] = p_L x - (1-p_L)T - c(L) \tag{2.2}$$

where $E[\pi_L]$ is the expected profit, p_L is the enhanced probability of evading detection when L layers are used, x is the total illicit funds, T is the total penalty if caught, and $c(L)$ represents the monotonic cost function associated with establishing and maintaining the L transaction layers. Because the fraud ring dramatically increases the probability of evading detection ($p_L \to 1$ as L increases), the expected payoff remains highly profitable even when factoring in the complex layering costs $c(L)$ [7].

2.3 MONEY LAUNDERING PATTERNS

Understanding the structural patterns that criminality assumes is fundamental to develop effective detection models and to employ strategies to

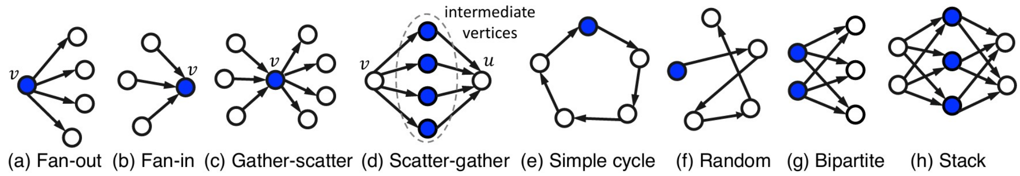

Figure 1.: Money Laundering patterns

enhance the explainability of them. According to the literature, there are 8 different patterns that are often used in ML [14, 2]. Which are:

- *fan-out*, shown in Figure 1 a, characterized by outgoing edges from a node v to more than 2 adjacent nodes;
- *fan-in*, Figure 1 b, v is connected to more than 2 nodes by incoming edges;
- *gather-scatter*, Figure 1 c, is the combination of a fan-in and a fan-out pattern on the same node v;
- *scatter-gather*, Figure 1 d, also composed by a fan-in and a fan-out pattern but on two different nodes, sharing the same intermediate neighbors;
- *simple-cycle*, Figure 1 e, characterized by a sequence of transactions that starts and ends at the same node;
- *random*, Figure 1 f, similar to the cycle but founds do not return to the original account;
- *bipartite*, Figure 1 g, moves founds from a set of original nodes to a set of target nodes, with no transactions between accounts of the same set;
- *stack*, Figure 1 h, extends the bipartite pattern by adding a second bipartite layer.

It's important to mention that in some cases some patterns can be found in both illicit and licit transactions [16]. Specifically, *fan-in* and *fan-out* are patterns that are also frequent in legitimate, every-day transactions, such as salary payments or transfers between family members or friends. That being said, the detection of such patterns remains a foundamental component of any money laundering detection models, albeit insufficient on its own. Furthermore, a model that correctly classifies those patterns provides a much more interpretable result than one that simply performs a binary classification between licit and illicit transactions. In Chapter 3 we will see why explainability is of fundamental importance in the financial field.

3

ETHICS, PRIVACY AND REGULATIONS

In this chapter we will begin by introducing a brief overview of the regulations that currently govern the field of AML. Subsequently a discussion on the main issues that the use of ML models may introduce will be provided, toghether with suggested solutions to address them. A special focus will be reserved to the issues of explainability and privacy. To conclude we will consider the use of Synthetic Datasets as a possible solution to these issues, analyzing their limits and advantages in the context of AML.

3.1 AML GLOBAL REGULATIONS

Effective ML detection requires a combination of domestic laws, regulatory oversight, and international cooperation. Since the end of the 20th century the United States and the Eureopean Union as key jurisdictions, toghether with the FAFT [1] as the global leading standard-setter have been developing and elaborate framework for AML [1].

In 1989 the G-7 nations enstabished an inter-governmental task force to develop and promote international guidelines in regard to fight money laundering, the FATF. Nowadays FAFT still holds a central role in the AML field. It has issued 40 recomendations for national laws and international cooperation [1]. These principles have been adopted by over than 200 jurisdictions and they fall into 5 main areas [12]:

- Risk Based Approach (RBA): jurisdictions must identify and understand the financial crime risks that they face, allocating the right amount of resources to mitigate them;

1 The Financial Action Task Force (FATF) is an intergovernmental policymaking body established in 1989 by the G7

- Legal Framework: members nations must adopt laws that criminalize money laundering; furthermore authorithies must be given the power to freeze and confiscate funds when needed;
- Preventive Measures:
  - Customer Due Diligence (CDD): financial instututions must verify the identity of their customers;
  - Ultimate Beneficial Ownership (OBO): company's owners must be verified, this adresses the issue of criminals hiding behind shell instututions;
  - Record Keeping: financial instututions must mantain their transactions recrds for at least 5 years and file Suspicious Transactions Reports (STRs) to Financial Intelligence Units (FIUs).
- Supervision and Transparency: regulators must actively supervise financial instututions; members nations must immediatly enforce UN financial sanctions when required to.
- International cooperation: members must quickly share information and provide mutual legal assistance in international investigations.

In regard to those guidelines FAFT keeps a *black-list* and a *grey-list* of jurisdictions that are not compliant with its standards [1].

3.2 ETHICAL AND PRACTICAL ISSUES

When considering real-world datasets, models that detect financial crime are trained on sensitive data that needs to be protected. To avoid the leakage of sensitive information, before feeding ML models, it is necessary to apply privacy-preserving techniques. The most common practice is to apply differential privacy, which adds noise to the dataset to protect individual data points. That noise can be quantified using the privacy budget ϵ, which determines the trade-off between privacy and utility. All this comes at the cost of reducing the accuracy of the model.

Another significant issue to consider is explainability. Regulators require banks to explain their decisions, and this requirement extends to the ML models they use. Deep Learning models, albeit generally more precise and powerful than traditional ML models, are often *black-boxes*, meaning their decision process is very difficoult to understand for a human being.

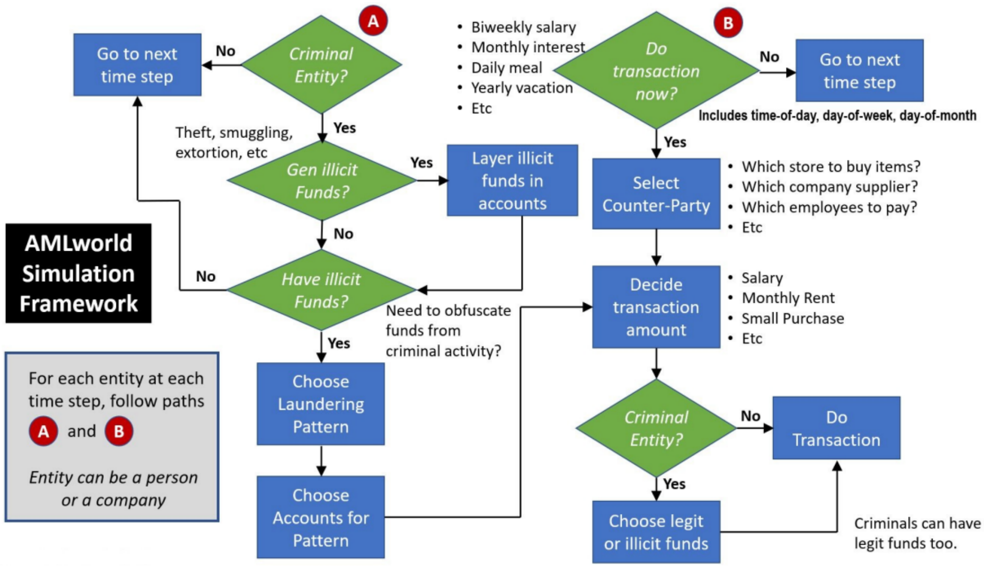

Figure 2.: Overview of AMLworld simulation [2]

Real datasets are also often limited by the fact that the informations are not shared accross jurisdictions, making it difficult to detect frauds that cross borders.

To tackle all those problematics researchers at IBM developed a simulator for generating realistic financial transactions, called AMLWorld (2023) [2]. This simulator solves the privacy problem, simulates multi-jurisdictional transactions and it even tries to takle the explainability issue by providing as lable not only a boolean value for the presence of a fraud, but also the ML pattern that caracterized it. This wouldn't be possible in real life, but models trained on this synthetic dataset could be able to generalize the detection of patterns even in real world scenarios, highly enhancing interpretability. Figure 2 provides an overview on how AMLWorld generates realistic financial transactions and patterns of financial crime.

The possible laundering patterns are the one described in Section 2.3; the model assumes the existence of 9 different surces of criminal activities: extortion, loan sharking, gambling, prostitution, kidnapping, robbery, embezzlement, drugs, and smuggling. The amount of money and the frequency of these criminal enterprises vary by both activity category and acting entity [2]. The funds are then inserted in the financial system alongside the legit ones.

4

TASK

In this dissertation we consider the problem of Edge-Level ML detection. This means, considering the Graph $G = (V, E)$ as in Definition 1.1, trying to learn a function f that, for each transaction $e \in E$, predicts its class. In particular we will consider two different settings. A binary classification where each edge is labeled as fraudulent or lecit and a multi-class classification where, in case of illicit transaction, we also predict the type of pattern, amongst the one described in Section 2.3, it's involved in.

Formally we seek a function f that minimizes the error $L(f)$ over a distribution $\mathcal{D}$ of Graphs.

$$\text{argmin}_{f \in \mathcal{H}} \left\{ L(f) \doteq \mathbb{E}_{(x,y) \sim \mathcal{D}}[l(f(x), y)] \right\} \tag{4.1}$$

Practically there's no way to know $\mathcal{D}$. Instead we will consider a dataset $G_D = (V_D, E_D)$, where both nodes and edges are equipped with feature vectors and we will seek a function f that minimizes $L(f)$ over this dataset computing the empirical risk:

$$\text{argmin}_{f \in \mathcal{H}} \left\{ \hat{L}(f) \doteq \frac{1}{|E_D|} \sum_{e \in E_D} l(f(x_e), y_e) \right\} \tag{4.2}$$

This is just the theoretical overview of the problem. In Chapter 5 we will provide more insights on how to define the hypothesis space $\mathcal{H}$ and the loss function l.

As mentioned in Chapter 3, the strict privacy regulations that governs the financial world, makes it hard to find real-world datasets. In this dissertation, for the reasons explained in Section 3.2, we will use the synthetic dataset *IBM AMLSim*. This is perfect for our scope, since it provides edge lables for both binary and multiclass classification.

	Small		Medium		Large	
Statistic	**HI**	**LI**	**HI**	**LI**	**HI**	**LI**
# of Days Spanned	10	10	16	16	97	97
# of Bank Accounts	515K	705K	2077K	2028K	2116K	2064K
# of Transactions	5M	7M	32M	31M	180M	176M
# of Laundering Transactions	3.6K	4.0K	35K	16K	223K	100K
Laundering Rate (1 per N Trans)	981	1942	905	1948	807	1750

Figure 3.: IBM AMLSim datasets statistics [2]

4.1 DATASET DESCRIPTION

The IBM AMLSim dataset comes in 6 different versions.

- HI-Small
- HI-Medium
- HI-Large
- LO-Small
- LO-Medium
- LO-Large

Where HI (High) and LO (Low) indicate the frequency of illicit transactions and the suffixes Small, Medium and Large indicate the size of the dataset. Figure 3 shows the differences between the 6 datasets.

In this work we will use the HI-Small version. Bigger datasets would of course garantee a better generalization and accuracy but, for hardware and time constraint, it would be impossible to train the models on such graphs. That begin said, we belive that the obtained results are good enough to understand how the models would perform in a real world scenario.

The HI-Small dataset is composed of 3 different files:

- `HI-Small_account.csv`, contains 518,581 accounts toghether with their features:
  - *Entity ID*: The ID of the account;
  - *Entity Name*: The name of the account;
  - *Bank Name*: The name of the bank holding the account;
  - *Bank ID*: The ID of the bank holding the account;

- *Account Number*: The number of the account;

Figure 4 shows some examples of account records.

- HI-Small_Trans.csv, contains transactions features and binary labels:
  - *Timestamp*: The timestamp of the transaction;
  - *From Bank*: The ID of the sending bank;
  - *From Account*: The ID of the sending account;
  - *To Bank*: The ID of the receiving bank;
  - *To Account*: The ID of the receiving account;
  - *Amount Received*: The amount of money received by the recipient;
  - *Receiving Currency*: The currency of the received amount (USD, EUR, etc.);
  - *Amount Paid*: The amount of money paid by the sender;
  - *Payment Currency*: The currency of the paid amount;
  - *Payment Format*: The format/method of the payment;
  - *Is Laundering*: Binary label indicating whether the transaction is illicit;
  - *src*: Source node identifier;
  - *dst*: Destination node identifier;

  Figure 5 shows some examples of transaction records.

- HI-Small_Patterns.txt, lists the edges that are involved in ML activities toghether with the type of pattern they are involved in (multiclass labels). Examples of the 8 differt types of ML patterns in IBM AMLSim are shown in Figure 6.

4.2 FEATURES ANALYSIS

In this section we provide a detailed analysis of the edges and nodes features distributions. Before diving into that we first take a look at the distribution of the labels in the dataset. There are 5,177 illicit transactions over a total of 5,078,345 Edges. Figure 7, provides the distribution of the multiclass label (type of pattern) over the fraudulent edges. It's important

	Bank Name	Bank ID	Account Number	Entity ID	Entity Name
0	Portugal Bank #4507	331579	80B779D80	80062E240	Sole Proprietorship #50438
1	Canada Bank #27	210	809D86900	800C998A0	Corporation #33520
2	UK Bank #33	21884	80812BE00	800C47F50	Partnership #35397
3	Germany Bank #4815	32742	81047F300	80096F0B0	Corporation #48813
4	National Bank of Harrisburg	127390	80BD8CF00	800FB8760	Corporation #889

Figure 4.: IBM AMLSim account records

	Timestamp	From Bank	From Account	To Bank	To Account	Amount Received	Receiving Currency	Amount Paid	Payment Currency	Payment Format	Is Laundering	src	dst
0	2022-09-01 00:20:00	010	8000EBD30	010	8000EBD30	3697.34	US Dollar	3697.34	US Dollar	Reinvestment	0	010_8000EBD30	010_8000EBD30
1	2022-09-01 00:20:00	03208	8000F4580	001	8000F5340	0.01	US Dollar	0.01	US Dollar	Cheque	0	03208_8000F4580	001_8000F5340
2	2022-09-01 00:00:00	03209	8000F4670	03209	8000F4670	14675.57	US Dollar	14675.57	US Dollar	Reinvestment	0	03209_8000F4670	03209_8000F4670
3	2022-09-01 00:02:00	012	8000F5030	012	8000F5030	2806.97	US Dollar	2806.97	US Dollar	Reinvestment	0	012_8000F5030	012_8000F5030
4	2022-09-01 00:06:00	010	8000F5200	010	8000F5200	36682.97	US Dollar	36682.97	US Dollar	Reinvestment	0	010_8000F5200	010_8000F5200

Figure 5.: IBM AMLSim transaction records

to notice that this information is provided only for $3,149$ out of the $5,177$ illicit transactions.

We now procede by analyzing the edge features. First, we notice that *Amount Received* and *Amount Paid* are almost identical. From Figure 8 we can see that fraudulent trasnactions have the tendency to be bigger than lecit ones.

Continuing our analysis on edge features, we notice that the most common currency is USD for both fraudulent and licit transactions; most of the currencies have a similar proportion distribution from both fraudulent and licit transactions. From Figure 9 we can see that there are two exceptions: Shekel, which is more prevalent in licit transactions, and Saudi Riyal, which is significantly more prevalent in fraudulent transactions. It's important to notice that the graph shows the proportion of each currency with respect to the total number of transactions in each class; Since there are way more licit than fraudulent transactions, having similar distributions means that, for every type of currency, licit transactions still represent the vast majority.

Considering now the payment formats, by looking at Figure 10 we notice that ACH transactions are more frequently used for fraudulent transactions compared to other methods.

The last edge feature is *Timestamp*. Since working with timestamps can be tricky, we decided to extract to categorical features representing the hour of the day and the day of the week (Figures 11). Normally we would

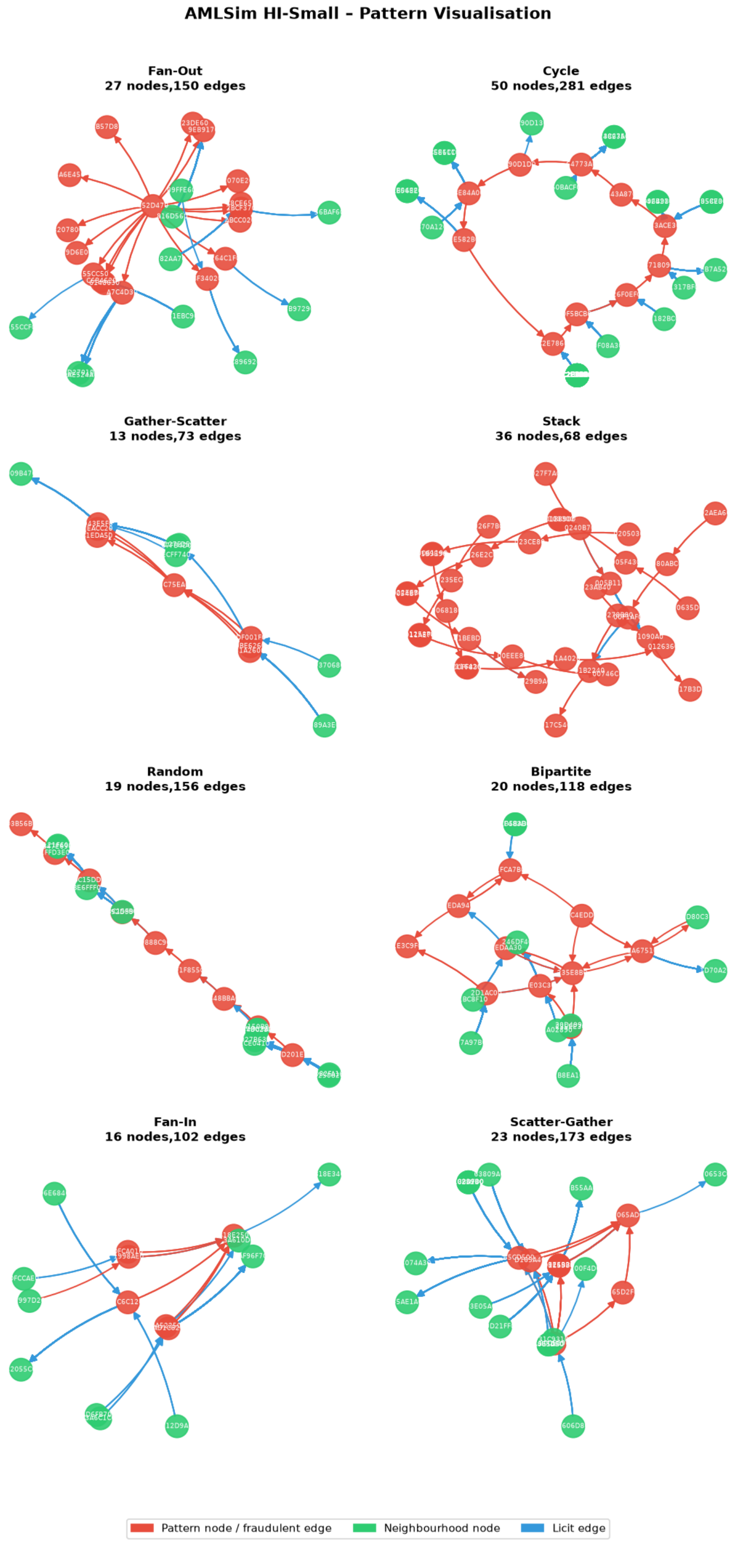

Figure 6.: IBM AMLSim patterns examples

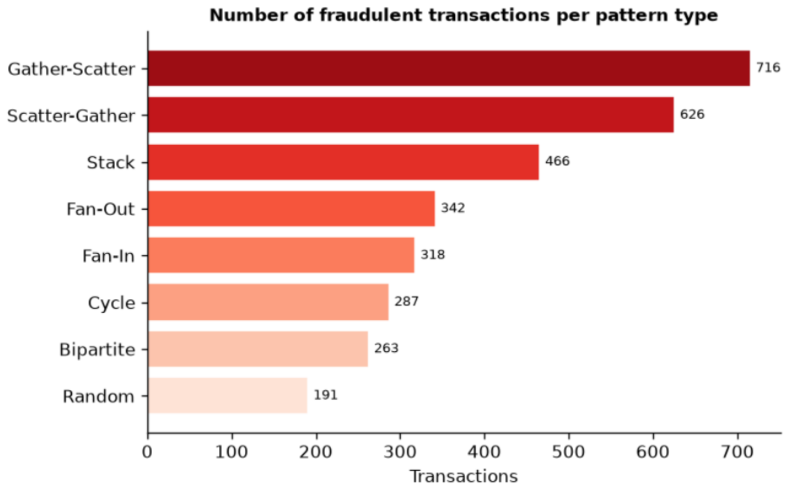

Figure 7.: IBM AMLSim patterns distribution over fraudulent edges

have also considered the week of the year and the year itself but, since IBM AMLSim Small only covers the span of a week, it would be irrelevant. That being said, it's important to mention that directly working with timestamps, albeit tricky, could lead to a more precise detection [Chapter 9].

Finally we take a look at the Node features. The only intresting one is *Bank Name*, Figure 12 shows the distribution of licit and illicit nodes over the 10 biggest banks.

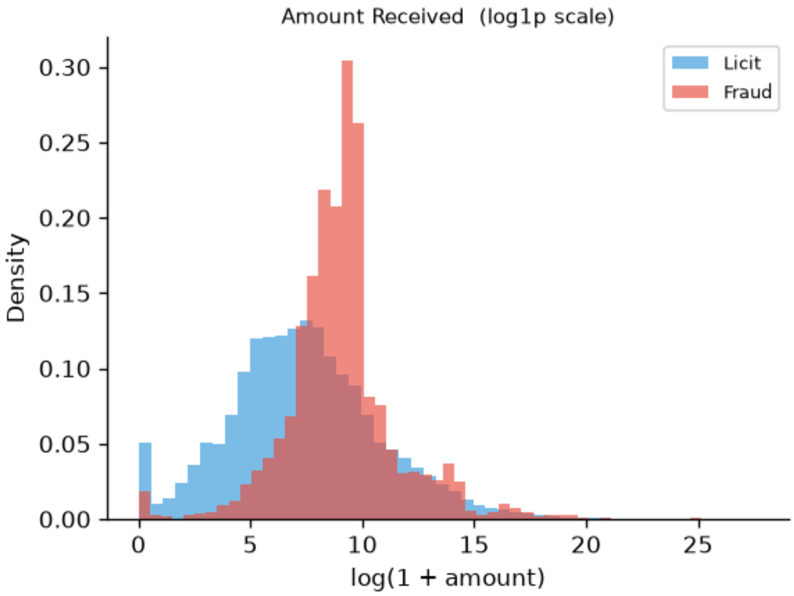

Figure 8.: IBM AMLSim amounts distribution

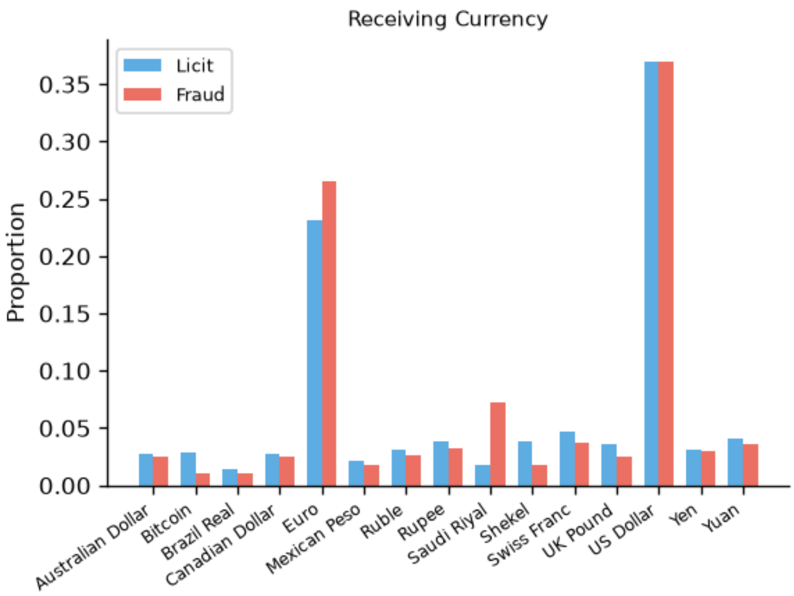

Figure 9.: IBM AMLSim currencies distribution

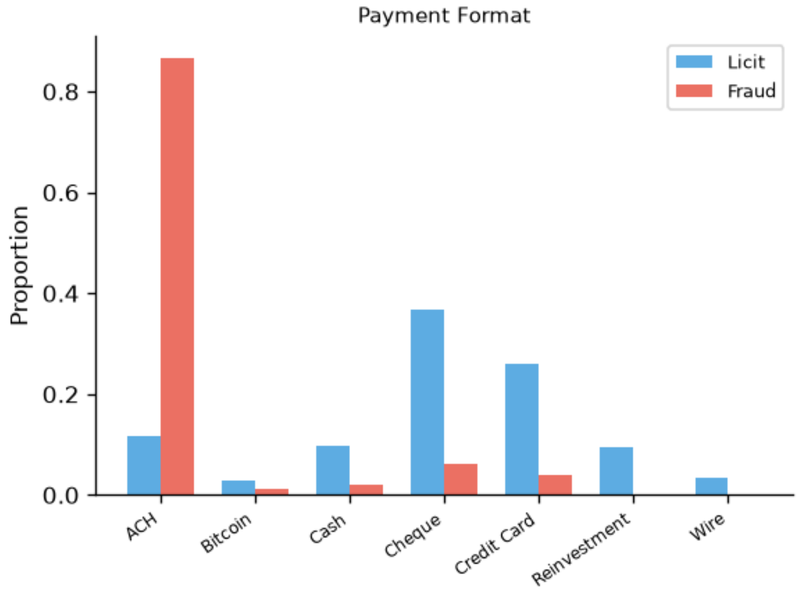

Figure 10.: IBM AMLSim payment formats distribution

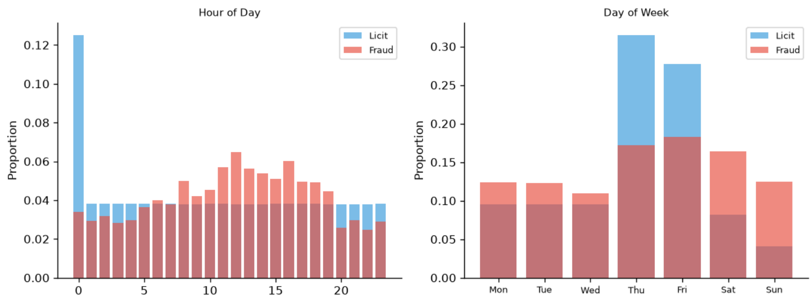

Figure 11.: IBM AMLSim hour and day of the week distributions

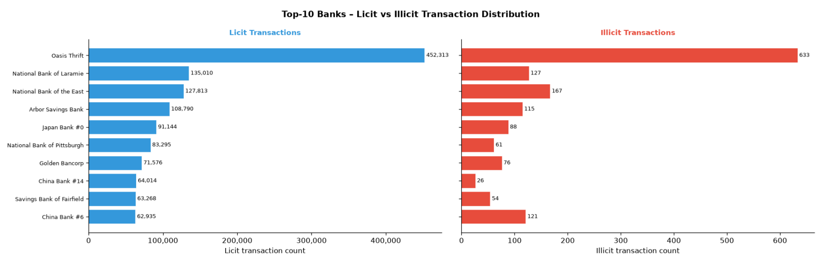

Figure 12.: IBM AMLSim banks distribution

5

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